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Advanced AI and ML Solution to Expedite Geological Rocktyping from Petrographic Thinsections to Improve Reservoir Characterization

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Abstract

Heterogenous carbonate reservoirs exhibit diverse flow characteristics. Petrographic study on thinsection images is essential for rock typing studies. Optical point counting is a conventional method which is semi-quantitative, and thinsection descriptions are in text format. Performing optical point counting is a cumbersome and time-consuming process requiring specialist skills and domain expertise. Optical point counting provides data for modal analysis to assess reservoir quality for rocktyping. Image processing of thinsection is a swift and robust alternative to optical point counting method. Proposed AI and ML solution automates modal analysis for digital reservoir description and augments available optical count data to large number of thinsection images in quick turnaround to enhance geological rocktyping.

An innovative AI and ML solution was developed by leveraging image processing algorithms and computing power to improve reservoir characterization. Computational algorithms mainly, multilevel thresholding and pixel intensity clustering algorithms were programmed to segment images and labelled by comparing with original image. The labeled elements were interpreted for geological elements such as matrix, pores, cement, and other granular content. Algorithms were customized for thinsection image processing by honoring complex pore types in carbonate reservoirs. The interpreted geological elements were measured for their physical properties like area, equivalent diameter, perimeter, solidity, eccentricity, and entropy. Geological rocktyping was carried out using K-Means classification based on digital proxy reservoir quality (DPRQ) parameters and compared with rock types made from conventional methods. The whole process was automated in a batch process for a specific reservoir type and computational cost was analyzed for optimization.

Automated modal analysis workflow produced digital thinsection data comparable to optical count data under microscopic analysis. Digital proxy rock quality (DPRQ) parameters were introduced and proved to be quick and reliable reference to expedite geological rocktyping instead of conventional workflows. Integration of laboratory nuclear magnetic resonance data enabled the bridge required for the integration of 2-D geometric properties with three-dimensional nature of conventional core analysis data from core plugs.

This paper demonstrated a unique and innovative solution to extract digital data from thinsections for modal analysis. Rocktyping workflow is enhanced over conventional workflow by defining DPRQ parameters. This paper presents a novel means to use thinsection images directly in digital format in

geoscience applications. As conventional workflows are limited by number of samples, time, cost and expert skill, this workflow can be scalable to large number of thinsections to expedite rocktyping in contemporary reservoir characterization workflows.

Introduction

Petrographic thinsection analysis provides comprehensive insights into geological constituents, pore characteristics, and micro scale heterogeneities. Optical point counting (Delesse 1848; Chayes 1949) remained a standard technique for quantitatively describing thinsections, though it requires meticulous observation by domain experts who systematically examine thinsection at various positions of microscope on the sample. This process, while informative particularly for modal analysis, can be labor-intensive due to challenges such as microscope navigation, grid selection, and the necessity of expert judgment and time. Modal analysis is usually undertaken by counting 300 points for both mineral identification and macroporosity. Point counting is carried out using an orthogonal grid system with equal distances between points (along 'x' and 'y' directions). The distance between points is variable depending on the thinsection area. Thinsection descriptions and modal data by subject matter expert is utilized for reservoir description to know depositional history and key diagenetic events. Geological rocktyping is conducted based on thinsection descriptions and modal data which consists of macro/micro porosity, cement and minerals identified. The challenge with this conventional workflow is extended work hours, limited samples and data format where descriptions are in text and semi-quantitative nature of modal data.

Advancements in image processing provided viable alternatives, addressing many limitations of conventional workflows to expedite rocktyping in reservoir characterization workflows. Petrographic Image Analysis (PIA) is a widely acknowledged method (Ehrlich and Korkowitz 1987; Ehrlich et al. 1991a) for image processing to analyze thinsection images for determining pore types. PIA provides two-dimensional geometric parameters of pore networks to generate three-dimensional petrophysical properties when there is an established correlation between 2-D measurements and 3-D core properties. The bridge between 2D measurements on image and 3D measurements at plug scale is a challenge in carbonate reservoirs due to varied and complex pore types. Bowers et al. (1995) and Carr et al. (1995) introduced a calibration protocol correlating NMR-derived pore types and image-derived classifications. Both approaches utilized NMR longitudinal relaxation time (T1) as a benchmark for validating image-based pore type identification which derives surface relaxivity (ρ) values that relate NMR T1 to pore size distributions. However, acquiring full NMR T1 distributions at logging speeds is not always practical. Alternatively, Reddy et al. (2022) presented a method based on NMR transverse relaxation time (T2) distributions, which also provide information on pore sizes under water-saturated conditions dominated by surface relaxation, are commonly measured both in laboratories and during well logging operations. Thus, calibrating image-derived pore types against NMR T2 distributions establishes a practical framework for estimating 2-D properties on thinsection images by honoring pore types in carbonates. A calibrated thinsection image can be further subjected to digital image processing to generate modal data such as matrix minerals, cement, 2D porosity and micro/macro pores. Derived modal data augmented existing modal data from conventional optical count method for deriving rock types.

Recent advancements in digital image processing, particularly through machine learning and artificial intelligence applications in computer vision, have further enhanced analytical capabilities. The paper applies modern image segmentation techniques—including multilevel thresholding (Chen and Chung 2001) and K-Means++ algorithm (Arthur and Vassilvitskii 2007) tailored specifically for complex carbonate reservoir thinsections. Integrating laboratory NMR data has facilitated effective linkage between 2-D geometric features and conventional core analysis. Optimization of computational processes has enabled automated batch processing of multiple samples, significantly accelerating digital reservoir characterization to expedite geological rocktyping.

Conventional workflow

Geological rocktyping forms the basis for reservoir rocktyping in reservoir models by integrating with petrophysical groups derived from MICP, RCA and well logs. Conventional workflow for geological rocktyping involves sedimentological core descriptions by domain experts followed by thinsection descriptions under microscope. The resulting data from domain experts is in text format with a depth wise record of observations on convention cores as a sedimentological log. Selected samples are cut on chips from core plugs to generate a photomicrograph of thinsection samples with vacuum impregnated by blue-dyed epoxy resin. Domain expert conducts petrographic study on photomicrographs under microscope for detailed thinsection description. Figure 1 shows a typical sedimentological log and presents a thinsection description for a core belonging to a heterogenous carbonate reservoir in an onshore field, Abu Dhabi. Target reservoir is dominantly a limestone with traces of dolomite. Figure 1 is illustrated with depth wise record of observation made on core in terms of lithofacies, grain types, pore types and other structural features. In the thinsection image of Figure 1, dark areas indicate granular and mineral content, blue indicates pores, and light color represents cement. Modal analysis results from conventional methods are also presented on the right side in Figure 1. Key parameters for each thinsection are derived as summary table, an example given below the thinsection image in Figure 1. Conventional study on all available thinsection images resulted in a comprehensive dataset for geological rocktyping.

Sedimentological and petrographic study integrating 286 thinsection samples and modal analysis on 32 selected samples identified four geological rocktypes in target reservoir. The characteristics of identified four rock types are given in Table 1 for the reservoir. Geological rocktyping confirmed a dominant mud supported limestones such pelloid packstone and skeletal wackestones represented by GRT2 to GRT 4. Grain supported limestones such as pelloid grainstone are less in extent in the reservoir represented by GRT1. Essential characteristics of the four geological rocktypes are given below.

- GRT1 was dominated by SPG and RPF lithofacies which were deposited in shallow subtidal to intertidal, high-energy, open platform. Less microporosity and more dissolution was observed from modal data
- GRT2 was dominated by SPP and BSPP lithofacies which were deposited in shallow subtidal to intertidal, moderate-energy restricted and open platform and inner shoal. moderate microporosity and less dissolution was observed from modal data
- GRT3 was dominated by SPWP and BSPWP lithofacies which were deposited in subtidal, low-energy open platform near fair weather wave base and inner shoal. High microporosity and high cementation was observed from modal data
- GRT4 was dominated by BBSW and WBSW lithofacies which were deposited in shallow subtidal, low energy restricted platform. Inner ramp, restricted lagoon. Compaction related effects and low porosity was observed from modal data

Available modal data on thinsection images of 32 samples was considered limited to characterize such a heterogeneous carbonate reservoir. Acquiring additional thinsection petrographic data was constrained by time, expert skill and budget. Therefore, it was essential to augment additional data through alternate methods on available thinsection images.

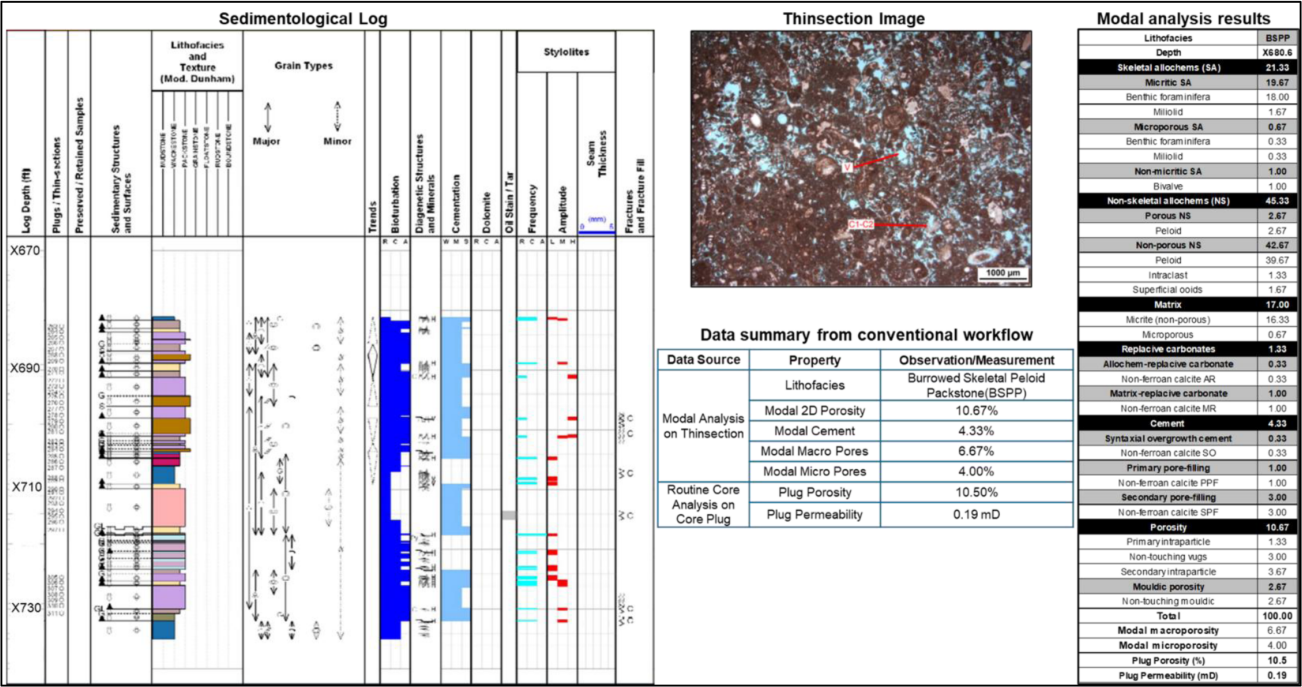


Figure 1—Typical sedimentological log and thinsection description in a reservoir through conventional workflow

Table 1—Characteristics of identified geological rocktypes in a reservoir through conventional workflow

Geological Rocktypes	Dominant Lithofacies	Lithofacies Description	Depositional Groups	Depositional Description	Key Diagenetic Events and percentage			
					Cementation	Dissolution	Dolomitisation	Microporosity
GRT1	SPG, RPF (Minor presence)	Skeletal pelloid Grainstone	Dep. Group 1 (patch-reef) Dep. Group 5 (lagoon to shoal)	Shallow subtidal to intertidal, high-energy , open platform. Upper ramp beach, near shoal crest and near inner shoal				
GRT2	SPP, BSPP	Skeletal Pelloid Packstone	Dep. Group 5 (lagoon to shoal)	Shallow subtidal to intertidal, moderate-energy restricted and open platform and inner shoal				
GRT3	SPWP, BSPWP, SW, BSW (Minor presence)	Skeletal Pelloid Wacke-Packstone	Dep. Group 5 (lagoon to shoal) & Dep. Group 6 (lagoon)	Subtidal, low-energy open platform near fair weather wave base and inner shoal				
GRT4	BBSW, WBSW, BBSP, WBSPP (Minor presence)	Burrowed Skeletal Wackestone	Dep. Group 7 (restricted lagoon)	Shallow subtidal, low energy restricted platform. Inner ramp, restricted lagoon				

Rocktype	Cement	Dissolution	Dolomitisation	Microporosity
GRT1	4.33%	6.67%	0.33%	10.67%
GRT2	4.33%	6.67%	0.33%	10.67%
GRT3	4.33%	6.67%	0.33%	10.67%
GRT4	4.33%	6.67%	0.33%	10.67%

AI and ML solution

As conventional microscopic analysis is repetitive and time consuming, the manual process limits the overall progress of rock quality assessment for geological rocktyping. Data format of the results is another challenge which is usually not readily compatible with conventional core analysis data for direct input into geoscience applications. An automated image processing (Reddy et al. (2022)) is an emerging solution to batch process thinsection images for digital reservoir description. Thinsection descriptions should be consistent, quantifiable, and correlated to conventional core analysis data for a robust reservoir characterization.

The solution leverages the modern image segmentation techniques namely multilevel thresholding (Chen and Chung 2001), K-Means++ algorithm (Arthur and Vassilvitskii 2007) and computing power. Algorithms were customized for thinsection image processing for images available in target reservoir by integration of laboratory nuclear magnetic resonance (NMR) which enabled the bridge required for the integration of 2-D geometric properties with conventional core analysis data. Batch processing of multiple thin sections paved the way for digital reservoir description in a swift turnaround time. Digital thinsection descriptions were used for generating equivalent modal data by validating the data with existing modal data. The solution was implemented on all other thinsection images where modal data was not available to augment overall database for geological rocktyping. Automated modal analysis workflow produced digital thinsection data

comparable to optical count data under microscopic analysis. Digital proxy rock quality (DPRQ) parameters were generated and proved to be quick and reliable reference to expedite geological rocktyping instead of conventional workflows

Theoretical Overview

Segmentation of thinsection image and estimation of geometric properties of constituents present in the image is the premise in image processing methods. The algorithm in the proposed solution is based on pixel intensity clustering algorithm for multilevel image segmentation. The brief overview is discussed here from relevant literature.

Augmented K-Means clustering for Multilevel Image Segmentation

Carbonate reservoirs undergo multiple paragenetic sequences developing complex pore types and matrix contents. Therefore, multilevel image segmentation such as Otsu method (Otsu 1979) is complicated for this application. K-Means clustering algorithm is simple and has a low computational cost. Clustering is an unsupervised learning method to define a finite set of clusters to classify pixels in an image. The color and pixel intensities in thinsection image are very heterogenous due to chemical compositions of minerals and tone changes due to stress induced deformation and diagenetic alternations. Augmented K-Means clustering is considered suitable due to the multi-dimensional nature of thinsection image. Prior knowledge of the number of clusters is required to classify pixel intensities based on intra-cluster similarity maximization and inter-cluster similarity minimization. Euclidean distance is computed to assign pixels to respective clusters in K-Means clustering method. Prior definition of number of cluster and initialization procedure is critical as improper initialization led to dead centers with no pixels.

In this solution, conventional K-Means algorithm is re-programmed in python for multilevel image segmentation. Firstly, the number of clusters is defined based on elbow method. In the elbow method, the variance which is Within-Cluster Sum of Squared errors (WCSS) is plotted against the number of clusters. The optimal number of clusters is found out from the elbow point on the plot of WCSS and number of clusters. With that, the number of clusters can be determined with reasonable accuracy. Secondly, dead centers of pixel clusters are avoided by careful seeding weighted by specific probability as proposed in K-Means++ algorithm while initialization procedure.

Workflow

The AI and ML solution was implemented in all available thinsection images of a heterogeneous carbonate reservoir. The workflow is illustrated in a flow chart presented in Figure 2. The workflow consisted of pre-processing, image processing and computation. Pre-Processing involved an initial geological review followed by image enhancement techniques to improve image pixels. The key steps of image processing are K-Means clustering, multilevel thresholding, image poretype calibration to NMR T2 and measurement of 2D geometric properties. Each of these steps are further discussed with an example from the results of thinsection image processing.

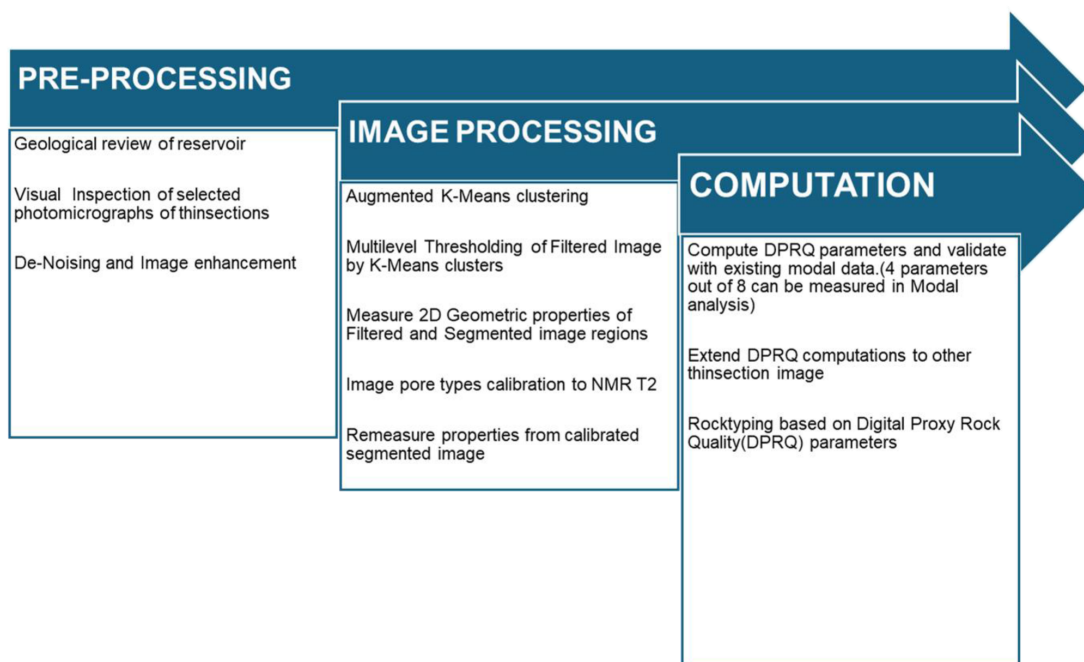


Figure 2—Workflow of AI and ML solution as implemented on thinsection images for digital thinsection data

Figure 3 shows an example of augmented K-Means clustering. K-Means clustering of image pixels resulted in 3 dominant clusters which were segmented by thresholding and labeled for geological constituents such as matrix, pores and cement. More heterogeneous image samples consisted of pixel clusters up to 4 clusters including micrite content or special minerals.

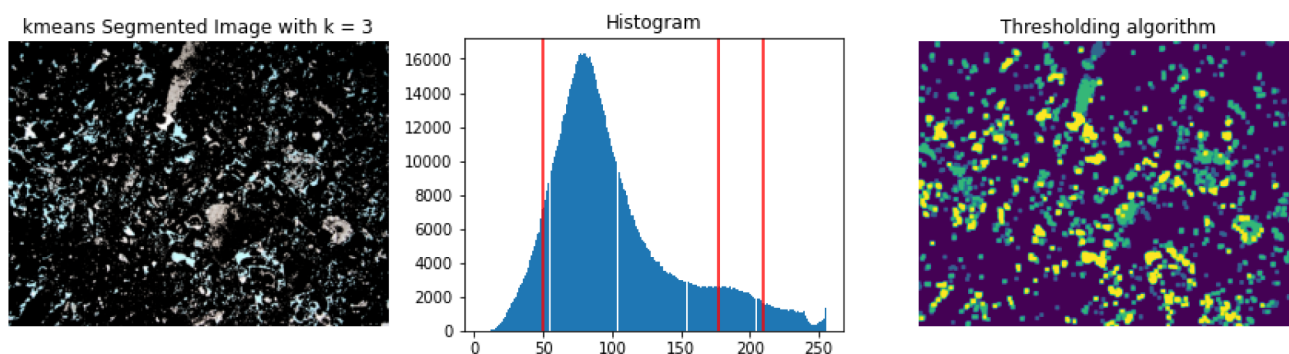


Figure 3—Thinsection image filtered by K-Means clustering and segmented by thresholding

Segmented images were further used for measurement of geometrical properties of individual labeled regions on the image. Digital data was extracted from segmented images by measuring 2D geometrical properties mainly label, area, perimeter, eccentricity, equivalent pore size, mean intensity, minimum intensity, maximum intensity, solidity and shannon entropy. Calibration of segmented image was prerequisite where extracted pore size must be correlated to known pore sizes measured from alternate source. This was carried out by comparing image derived 2D pore sizes with Laboratory NMR T2 distribution. The difference between these two is a scaling coefficient when multiplied on 2D pore size data matched to NMR T2 distribution data. Theoretically the scaling coefficient is equivalent to the surface relaxivity of pore surface as per fundamentals of nuclear magnetic resonance and transverse relaxation time(T2). The estimated pore size from thinsection image was compared with lab measured NMR T2 distribution on core plug where thinsection sample was taken. Since laboratory NMR T2 is not available near to all

thinsection samples, the calibration was performed for all samples where laboratory NMR was available to know the scaling coefficient. Figure 4 presented an example for calibration of segmented image before and after calibration where image derived 2D pore types was scaled to NMR T2 derived pore types. An average scaling coefficient of 0.7 was observed from the calibration exercises on all image samples where an associated core plug was available with laboratory NMR T2 data. Calibration was performed on all other image samples belonging to the target reservoir using an average scaling coefficient. Calibrated images were then used to regenerate 2D geometrical properties for each thinsection image of the target reservoir.

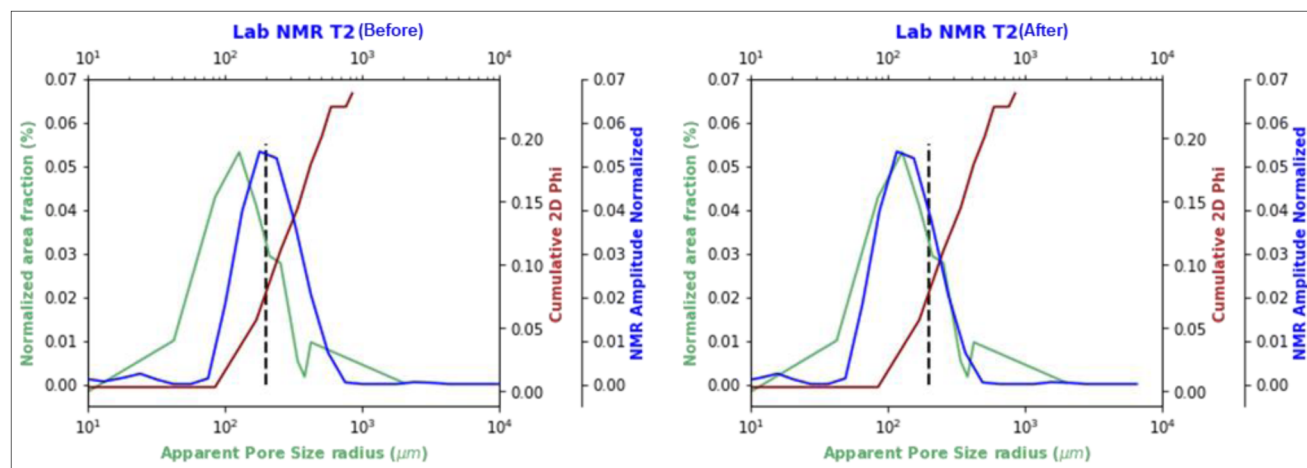


Figure 4—Calibration of segmented image before and after where image derived pore size is matched to NMR T2 on adjacent core plug

The resulting 2D geometric data was digital thinsection data which was an excel file in digital format for direct input to geoscience applications such as rocktyping. Key inputs to geological rocktyping are rock quality controls which contribute to rock permeability and flow capacity of reservoir. Digital Proxy Rock Quality (DPRQ) parameters were defined from digital thinsection data which are a proxy to understand pore enhancement and pore destruction due to diagenetic events. Conventionally thinsections are the source to understand reservoir heterogeneity. In the proposed solution, reservoir heterogeneity can be represented and captured in the form of DPRQ parameters in the digital thinsection description.

Eight DPRQ parameters namely- iPhi, iMacro, iMicro, iCement, shannon entropy, average poresize, average eccentricity, average perimeter were derived from digital thinsection data for each thinsection image of target reservoir. DPRQ parameters are a proxy to resolve reservoir heterogeneity quantitatively for geological rocktyping. Four of the eight DPRQ parameters iPhi, iMacro, iMicro and iCement, were also conventionally available from optical point counting in modal analysis. Validation was carried out for these four DPRQ parameters by comparing DPRQ parameters from AI-ML solution with the same parameters available in modal data. Figure 5 confirmed a good correlation between AI-ML results and conventional modal analysis results. In Figure 5, red colored points indicated a higher correlation score whereas blue colored points indicated a lower correlation score between both methods. Overall correlation score for each of the four DPRQ parameters were shown in Figure 6 validating reliability of AI-ML solution. Conventional optical point count method involved microscopic analysis, and AI-ML solution is computer vision processing, 45 to 50% score for iMicro and iCement is deemed good due to the difference in approach of both methods. Although validation was possible for only four DPRQ parameters from conventional method, extracting remaining four DPRQ parameters from the same calibrated image resulted in additional rock quality controls to improve geological rocktyping.

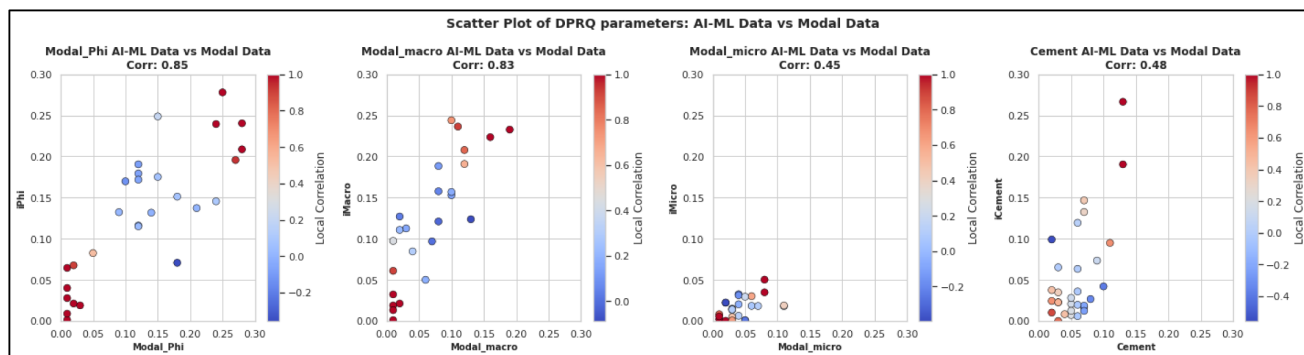


Figure 5—Validation of four DPRQ parameters from AI-ML solution with that of conventional modal data indicating correlation score for each point on color scale

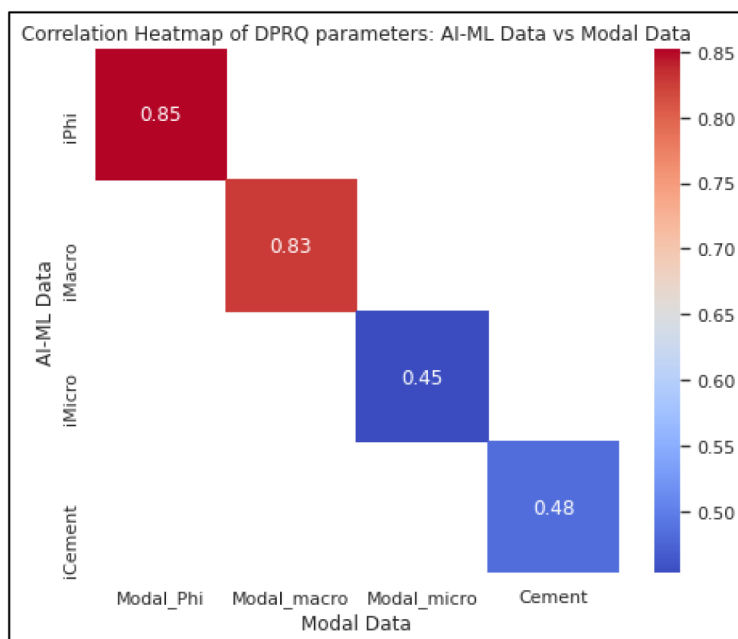


Figure 6—Overall correlation score for four DPRQ parameters from AI-ML solution with that of conventional modal data

Next step in AI-ML solution was to extend workflow to generate DPRQ parameters from all available thinsection images of target reservoir. The results were augmented existing modal data to substantiate thinsection interpretations to large number of samples in a consistent, quantifiable and in swift turnaround. A scatter plot of permeability measurements on core plug with each of eight DPRQ parameter from thinsection image were presented in Figure 7 for both original modal samples and additional samples where AI-ML solution was implemented. The scatter plot of each DPRQ parameter with permeability denoted a reasonable correlation trend such as correlation was apparently positive trend with permeability for iMacro and apparently negative trend for iCement with permeability. Eight DPRQ parameters were found to be the digital proxy to resolve and capture reservoir heterogeneity to perform geological rocktyping. K-Means classification was performed with the input of eight DPRQ variables to identify and predict dominant geological rocktypes in target reservoir. The unsupervised classification based on elbow method to detect the minimum rock classes resulted in four geological rocktypes(iGRT) in target reservoir. Predicted geological rocktypes(iGRT1, iGRT2, iGRT3 & iGRT4) by AI-ML solution were compared with the geological rocktypes from conventional methods and discussed in next section.

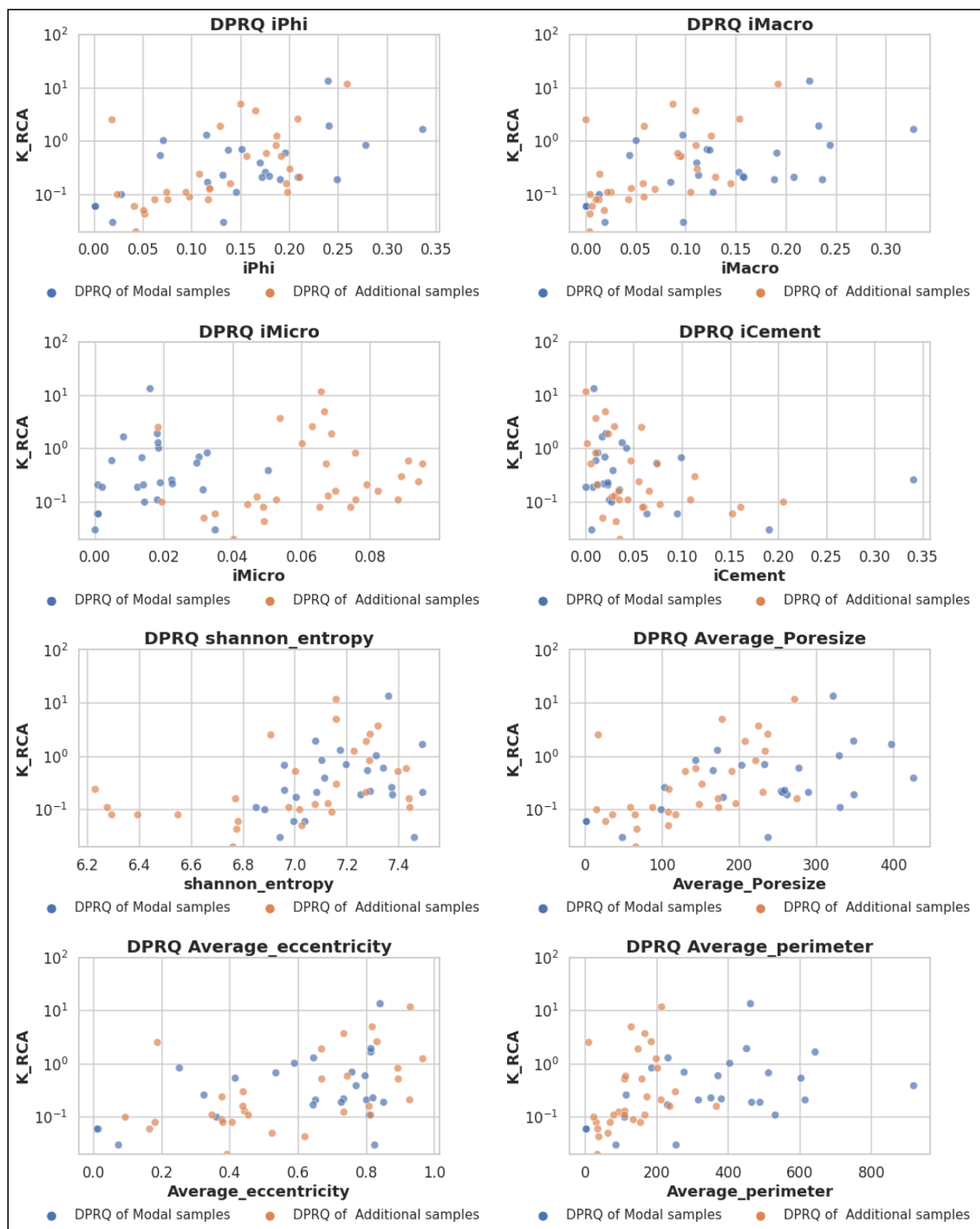


Figure 7—Permeability measurements on core plugs versus DPRQ parameters of original modal samples and that of additional samples by extending DPRQ computations

Results and Discussion

AI-ML solution resulted in digital thinsection description by calibrating thinsection images to honor image derived pore types with NMR T2 on adjacent core plugs. A cross-validation was achieved to relate image derived 2D properties with adjacent 3D core plug measurements to enable digital thinsection data for

application in reservoir characterization workflows. Figure 8 presented a pairplot depicting four geological rocktypes and their variability in terms of eight DPRQ parameters which is the digital proxy to reservoir heterogeneity.

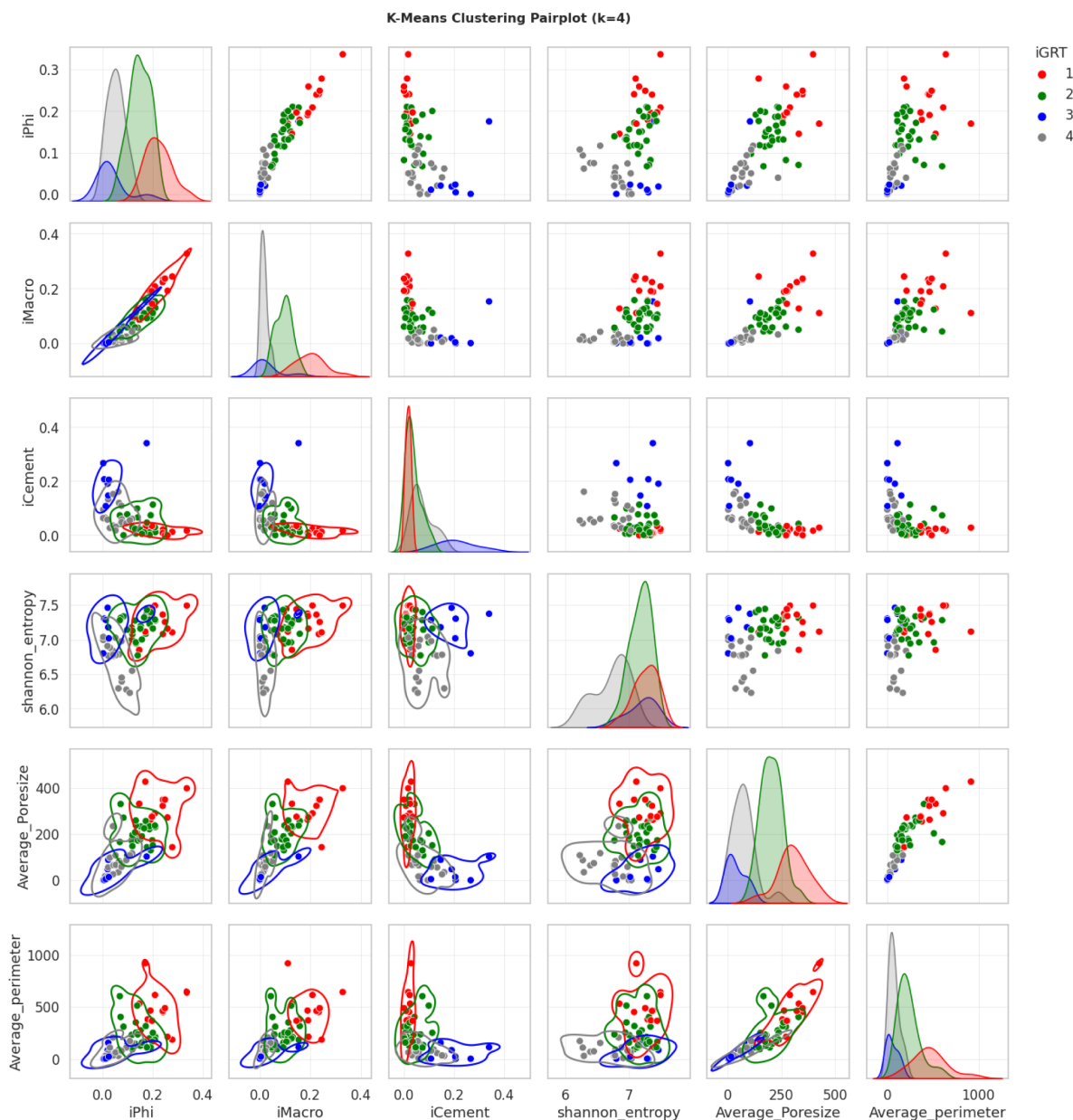


Figure 8—Pairplot of eight DPRQ parameters colored by geological rocktypes identified from AI-ML solution

Four geological rocktypes from AI-ML solution were summarized as follows.

- iGRT1 was less abundant and has the highest rock quality in the reservoir as represented by DPRQ parameters in pair plot of Figure 8. High porosity, high macro porosity, higher pore sizes and less cement was observed on images. High entropy is indicated by a mixture of tight matrix pores and diagenetically enhanced macropores.
- iGRT2 was very abundant and has the better rock quality in the reservoir as represented by DPRQ parameters in pair plot of Figure 8. Moderate DPRQ parameters were observed on images
- iGRT3 was very abundant and has moderate rock quality in the reservoir as represented by DPRQ parameters in pair plot of Figure 8. Poor to moderate DPRQ parameters were observed on images

- iGRT4 was less abundant and has poor rock quality in the reservoir as represented by DPRQ parameters in pair plot of Figure 8. Poor DPRQ parameters were observed on images

A comparison was made between iGRT from AI-ML solution and GRT from conventional method. Figure presented a heat map showing a contingency table between GRT and iGRT. 86% of Cramer's V and 83% of contingency coefficient was achieved between the results of both methods.

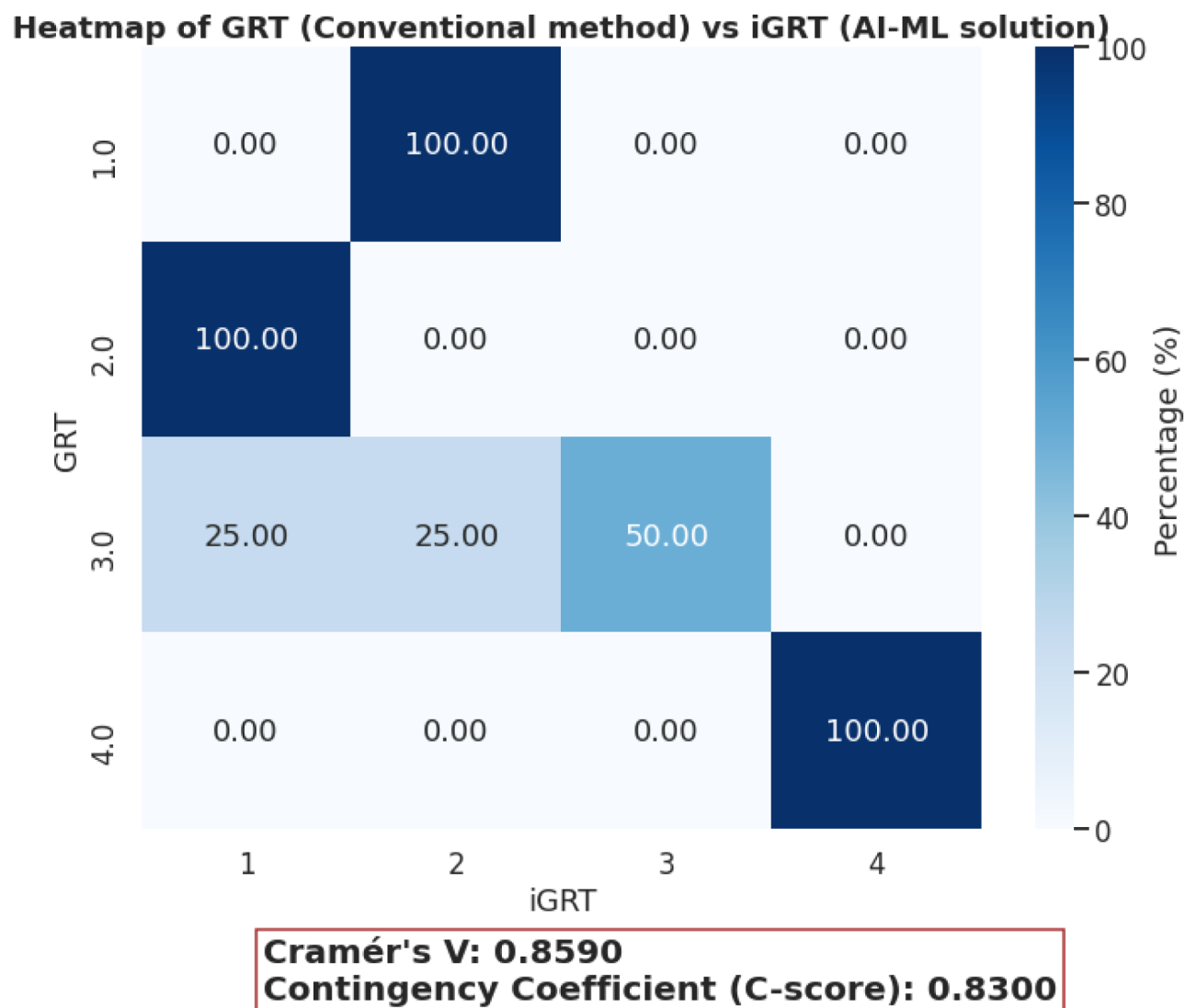


Figure 9—Heatmap of GRT (Conventional method) versus iGRT(AI-ML solution) as per contingency table

AI-ML solution was automated in a subroutine to analyze computational cost and execution time. Figure 10 presented total execution time to process the number of images with different pixel sizes in one minute of time frame. It was noted that a maximum of five images can be processed and digital thinsection data can be generated. The pre-requisite before scaling up the solution was that the subroutine must be calibrated and optimized to the specific reservoir type. Initial pre-processing and image pore type calibration to NMR T2 was an iterative process whose time is not included in total execution time of Figure 10.

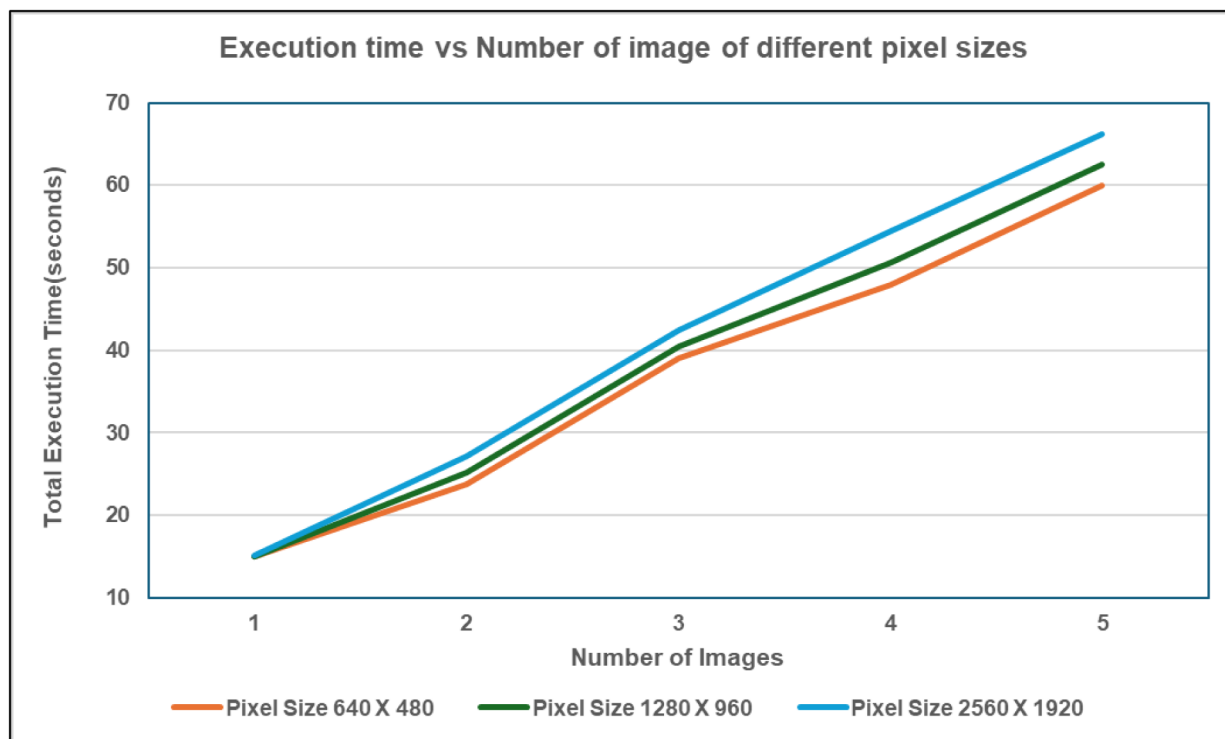


Figure 10—Execution time vs Number of image of different pixel sizes

Conclusions

The integration of advanced AI and machine learning solutions has demonstrated significant potential in expediting geological rock typing from petrographic thinsections. This innovative approach not only enhanced the robustness and efficiency of reservoir characterization workflow but also provided a more comprehensive understanding of subsurface geology. Key outcomes and merits of the solution were as follows:

- AI-ML solution is proven technique leveraging image processing and computing power to digitize thinsection interpretations
- Automated image processing offers a batch solution for digital reservoir description, enabling consistent, quantifiable thinsection analysis that correlates with conventional core analysis for robust reservoir characterization.
- As conventional microscopic analysis is repetitive and time-consuming, digital reservoir description from thinsection images expedited rock quality assessment by applying to large number of samples in quick turnaround.
- DPRQ parameters were introduced through the workflow which can be extracted as digital proxy to capture reservoir heterogeneity for geological rocktyping
- Geological rocktyping workflow is significantly enhanced and expedited to predict rocktypes using DPRQ parameters extracted from thinsection images
- The results of the solution were validated by existing conventional data and extended to additional sample images to fill data gaps in thinsection studies

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Nomenclature

AI-ML	= Artificial Intelligence and Machine Learning
GRT	= Geological Rocktype
iGRT	= Geological Rocktype from AI-ML
NMR	=Nuclear Magnetic Resonance
T2	= NMR transverse relaxation time[ms]
T1	=NMR longitudinal relaxation time[ms]
MICP	=Mercury injection capillary pressure
RCA	= Routine Core Analysis
ρ	= Coefficient of surface relaxivity

References

- Bowers, M.C., Ehrlich, R., Howard, J.J. and Kenyon, W.E. 1995. Determination of porosity types from NMR data and their relationship to porosity types derived from thin section *Journal of Petroleum Science and Engineering*, Volume **13**, Issue 1, Pages 1–14, ISSN 0920-4105, DOI: [https://doi.org/10.1016/0920-4105\(94\)00056-A](https://doi.org/10.1016/0920-4105(94)00056-A)
- Carr, M.B., Ehrlich, R., Bowers, M.C., Howard, J.J. 1995. Correlation of porosity types derived from NMR data and thin section image analysis in a carbonate reservoir. *Journal of Petroleum Science and Engineering*, Volume **14**, Issues 3–4, Pages 115–131, ISSN 0920-4105. DOI: [https://doi.org/10.1016/0920-4105\(95\)00045-3](https://doi.org/10.1016/0920-4105(95)00045-3)
- Chayes F., A. 1949. Simple point counter for thin section analysis, *American Mineralogist*. Volume **34**. Pages 1–11.
- Delesse M. A. 1848. Procédé mécanique pour déterminer la composition des roches, *Annales des Mines*. Volume **13**. Pages 379–388.
- David Arthur and Sergei Vassilvitskii. 2007. K-means++: the advantages of careful seeding. In *Proceedings of the eighteenth annual ACM-SIAM symposium on Discrete algorithms (SODA '07)*. Society for Industrial and Applied Mathematics, USA, 1027–1035.
- Ehrlich, R., and Horkowitz, J.P. 1987. Estimation of petrophysics from thin section: petrographic image analysis. *AAPG*, p. 238.
- Ehrlich, R., Crabtree, J., Horkowitz, K. and Horkowitz, J. 1991a. Petrography and Reservoir Physics I: Objective Classification of Reservoir Porosity. *AAPG Bulletin* **75** (10). 1547–1562. DOI: <https://doi.org/10.1306/0C9B2989-1710-11D7-8645000102C1865D>
- Liao, P., Chen, T., & Chung, P.C. 2001. A Fast Algorithm for Multilevel Thresholding. *Journal of Information Science and Engineering*, **17**, 713–727.
- Otsu, N. 1979. A threshold selection method from gray level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, **9**, 62–66.
- Rathnakar Reddy, Voleti, Deepak Kumar, Dutta, Ashim, Vanam, Prabhakar Reddy, and Nagaraju Reddicharla. "Automated Image Processing of Petrographic Thin Sections for Digital Reservoir Description: A Bridge to Correlate with Core and NMR Data." Paper presented at the ADIPEC, Abu Dhabi, UAE, October 2022. doi: <https://doi.org/10.2118/211691-MS>